



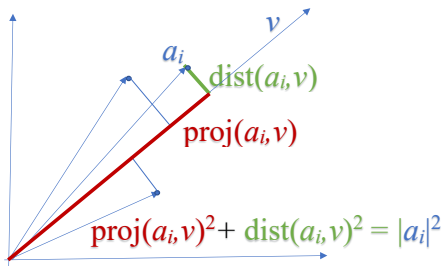
Advanced Algorithms

COMP4121

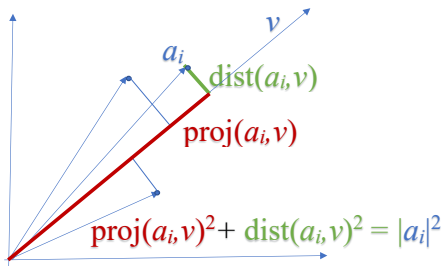
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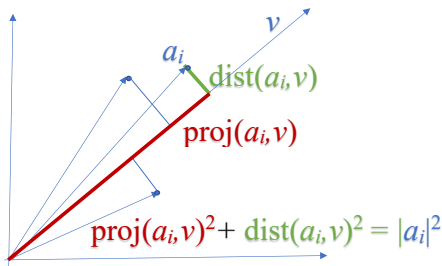
A brief sketch of the Singular Value Decomposition (SVD)



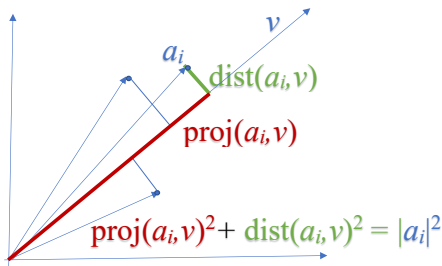
- Assume we are given n points $\{\mathbf{a}_i : 1 \leq i \leq n\}$ and wish to find a line, given by a vector $\mathbf{v}$, which is the closest to these points, in the sense that the sum of the squares of distances $\sum_{i=1}^n \text{dist}(\mathbf{a}_i, \mathbf{v})^2$ from these points to that line is minimal.
- Note that the orthogonal projections $\text{proj}(\mathbf{a}_i, \mathbf{v})$ of these points onto the line containing $\mathbf{v}$ satisfy $\text{proj}(\mathbf{a}_i, \mathbf{v})^2 + \text{dist}(\mathbf{a}_i, \mathbf{v})^2 = |\mathbf{a}_i|^2$.
- Thus, $\sum_{i=1}^n \text{dist}(\mathbf{a}_i, \mathbf{v})^2 = \sum_{i=1}^n |\mathbf{a}_i|^2 - \sum_{i=1}^n \text{proj}(\mathbf{a}_i, \mathbf{v})^2$.
- Since $\sum_{i=1}^n |\mathbf{a}_i|^2$ is constant, minimising $\sum_{i=1}^n \text{dist}(\mathbf{a}_i, \mathbf{v})^2$ is equivalent to maximising $\sum_{i=1}^n \text{proj}(\mathbf{a}_i, \mathbf{v})^2$.



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- The same applies not only to lines but also to subspaces: minimising the sum of squares of distances of a set of points to a subspace is equivalent to maximising the sum of the squares of the projections of the vectors corresponding to these points onto such a subspace.
- Assume again that we are given n points in a d dimensional space by vectors $\mathbf{a}_i$ corresponding to these points.
- We can now form a matrix $\mathbf{A}$ whose rows are the transposes $\mathbf{a}_i^T$ of the vectors $\mathbf{a}_i$ corresponding to these points.
- Let $\mathbf{v}$ be an arbitrary vector of unit norm, $|\mathbf{v}| = 1$; then

$$\sum_{i=1}^n \text{proj}(\mathbf{a}_i, \mathbf{v})^2 = \sum_{i=1}^n \langle \mathbf{a}_i, \mathbf{v} \rangle^2 = |\mathbf{A}\mathbf{v}|^2$$

- Thus, the best fit line is given by a vector $\mathbf{v}$ for which $|\mathbf{A}\mathbf{v}|^2$ is maximal.
- We define *the first singular vector of $\mathbf{A}$* to be vector $\mathbf{v}_1 = \arg \max_{|\mathbf{v}|=1} |\mathbf{A}\mathbf{v}|$.
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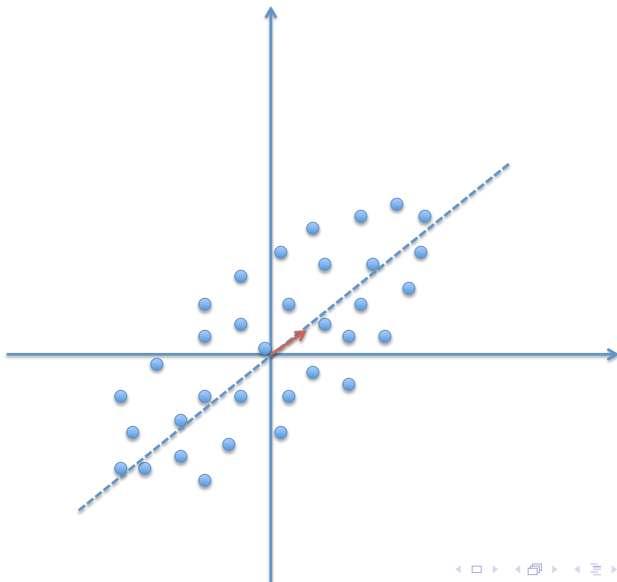
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SVD



- We now define $\mathbf{v}_2 = \arg \max_{|\mathbf{v}|=1; \mathbf{v} \perp \mathbf{v}_1} |\mathbf{A}\mathbf{v}|$ and $\sigma_2(\mathbf{A}) = |\mathbf{A}\mathbf{v}_2|$, and by recursion

$$\mathbf{v}_k = \arg \max_{|\mathbf{v}|=1; \mathbf{v} \perp \{\mathbf{v}_1, \dots, \mathbf{v}_{k-1}\}} |\mathbf{A}\mathbf{v}|;$$

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- The recursion stops when we reach r satisfying

$$\max_{|\mathbf{v}|=1; \mathbf{v} \perp \{\mathbf{v}_1, \dots, \mathbf{v}_r\}} |\mathbf{A}\mathbf{v}| = 0,$$

i.e., when all $\mathbf{v}$ such that $\mathbf{v} \perp \{\mathbf{v}_1, \dots, \mathbf{v}_r\}$ belong to the set of all $\mathbf{x}$ such that $\mathbf{A}\mathbf{x} = 0$, called *the kernel* of $\mathbf{A}$.

- **Theorem:** For all $1 \leq k \leq r$, the subspace V_k spanned by vectors $\mathbf{v}_1, \dots, \mathbf{v}_k$ is the subspace which minimises the sum of the squares of distances of points given by vectors corresponding to the rows of $\mathbf{A}$ to a k dimensional subspace, and maximizes sum of squares of orthogonal projections of these vectors to that subspace.

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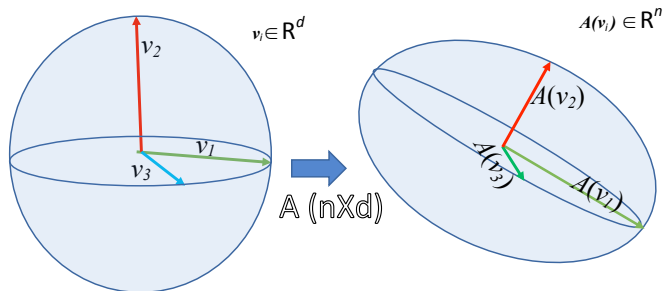
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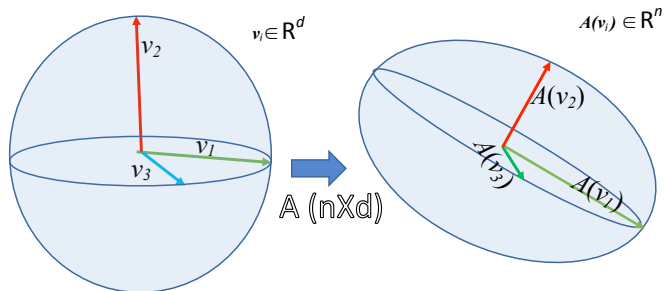
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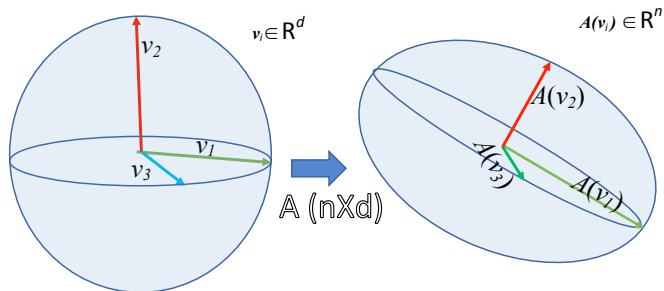
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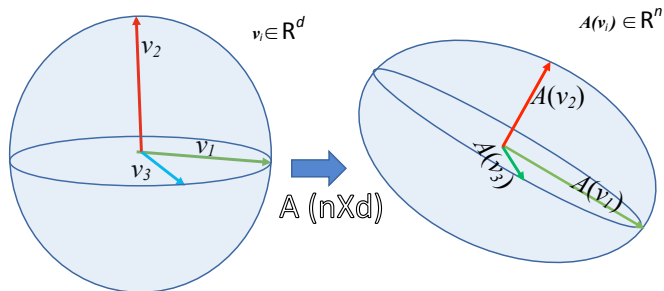
- Since vector $\mathbf{v}$ is such that $|\mathbf{A}\mathbf{v}|$ is maximal, $\mathbf{v}_1$ is the unit vector in the direction in which the unit ball $\{\mathbf{v} : |\mathbf{v}| = 1\}$ gets maximally elongated.
- Among all unit vectors orthogonal to $\mathbf{v}_1$, $\mathbf{v}_2$ is the unit vector in a direction in which the unit ball gets maximally elongated, and so on; see the figure.
- Note that vectors $\mathbf{v}_i$ on the left belong to a d dimensional space; vectors $\mathbf{A}(\mathbf{v}_i)$ on the right belong to an n dimensional space (n is the number of rows in $\mathbf{A}$) but the subspace they span is r dimensional where r can be smaller than the number n of rows of $\mathbf{A}$; thus the image of the unit ball might be of lower dimension than n , if the rank of matrix $\mathbf{A}$ is smaller than n .
- One can prove that vectors $\mathbf{A}\mathbf{v}_i$ are also orthogonal.



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- Let us represent each row vector $\mathbf{a}_j$ via its projections onto the unit vectors $\mathbf{v}_i$, $1 \leq i \leq r$, plus a component in a direction $\mathbf{x}$ orthogonal to all $\mathbf{v}_i$:

$$\mathbf{a}_j = \sum_{i=1}^r \langle \mathbf{a}_j, \mathbf{v}_i \rangle \mathbf{v}_i + \langle \mathbf{a}_j, \mathbf{x}_j \rangle \mathbf{x}_j$$

where $\mathbf{x}_j$ is a unit vector orthogonal to all $\mathbf{v}_i$, $1 \leq i \leq r$.

- Since for all $\mathbf{v}$ which are orthogonal to all $\mathbf{v}_1, \dots, \mathbf{v}_r$ we have $\mathbf{A}\mathbf{v} = \mathbf{0}$, each $\mathbf{x}_j$ satisfies $\langle \mathbf{a}_j, \mathbf{x}_j \rangle = 0$.
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$$|\mathbf{a}_j|^2 = \sum_{i=1}^r \langle \mathbf{a}_j, \mathbf{v}_i \rangle^2$$

- Adding up these equalities for all $\mathbf{a}_j$, $1 \leq j \leq n$ produces

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where $\mathbf{x}_j$ is a unit vector orthogonal to all $\mathbf{v}_i$, $1 \leq i \leq r$.

- Since for all $\mathbf{v}$ which are orthogonal to all $\mathbf{v}_1, \dots, \mathbf{v}_r$ we have $\mathbf{A}\mathbf{v} = \mathbf{0}$, each $\mathbf{x}_j$ satisfies $\langle \mathbf{a}_j, \mathbf{x}_j \rangle = 0$.
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$$|\mathbf{a}_j|^2 = \sum_{i=1}^r \langle \mathbf{a}_j, \mathbf{v}_i \rangle^2$$

- Adding up these equalities for all $\mathbf{a}_j$, $1 \leq j \leq n$ produces

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$$\sum_{j=1}^n |\mathbf{a}_j|^2 = \sum_{j=1}^n \sum_{i=1}^d A_{ji}^2$$

- This implies that the *Frobenius norm* of a matrix, namely the square root of the sum of squares of all of its entries, is equal to the square root of the sum of all of its singular values, i.e.,

$$|A|_F = \sqrt{\sum_{i=1}^n \sum_{j=1}^d A_{ij}^2} = \sqrt{\sum_{i=1}^r \sigma_i^2}$$

- Vectors $\mathbf{v}_i$, $1 \leq i \leq r$ are called the right singular vectors.
- Vectors

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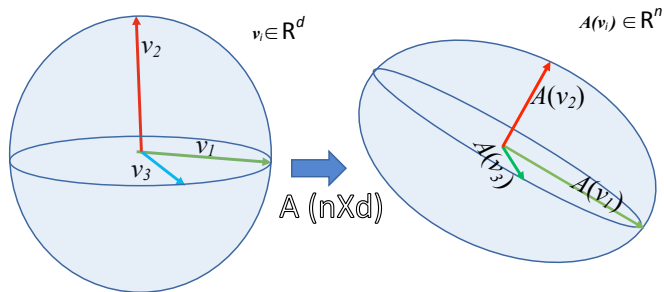
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- Matrices $\sigma_i \mathbf{u}_i \mathbf{v}_i^T$ are called the *principal components* of matrix $\mathbf{A}$.

• **Theorem:** $\mathbf{A} = \sum_{i=1}^r \sigma_i \mathbf{u}_i \mathbf{v}_i^T$

• **Proof:**

- It is enough to show that for all $\mathbf{x}$

$$\mathbf{A}\mathbf{x} = \sum_{i=1}^r \sigma_i \mathbf{u}_i \mathbf{v}_i^T \mathbf{x}$$

- But every vector $\mathbf{x}$ can be expressed as a linear combination of $\mathbf{v}_k$ plus a vector $\mathbf{y}$ which is orthogonal to all vectors $\mathbf{v}_k$

$$\mathbf{x} = \sum_{i=1}^r c_i \mathbf{v}_i + \mathbf{y}.$$

- Thus,

$$\mathbf{A}\mathbf{x} = \left(\sum_{i=1}^r c_i \mathbf{A}\mathbf{v}_i \right) + \mathbf{A}\mathbf{y}$$

- Since $\mathbf{y}$ is orthogonal to all vectors $\mathbf{v}_k$, it satisfies $\mathbf{A}\mathbf{y} = \mathbf{0}$ and thus

$$\mathbf{A}\mathbf{x} = \sum_{i=1}^r c_i \mathbf{A}\mathbf{v}_i = \sum_{i=1}^r c_i \sigma_i \mathbf{u}_i$$

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- **Proof continued:**

- Similarly,

$$\begin{aligned}
 \sum_{i=1}^r \sigma_i \mathbf{u}_i \mathbf{v}_i^\top \mathbf{x} &= \sum_{i=1}^r \sigma_i \mathbf{u}_i \mathbf{v}_i^\top \left(\sum_{j=1}^r c_j \mathbf{v}_j + \mathbf{y} \right) \\
 &= \sum_{i=1}^r \sum_{j=1}^r \sigma_i c_j \mathbf{u}_i \mathbf{v}_i^\top \mathbf{v}_j + \sum_{i=1}^r \sigma_i \mathbf{u}_i \mathbf{v}_i^\top \mathbf{y} \\
 &= \sum_{i=1}^r \sigma_i c_i \mathbf{u}_i
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- Thus we got

$$\mathbf{Ax} = \sum_{i=1}^r c_i \sigma_i \mathbf{u}_i \quad \text{and} \quad \sum_{i=1}^r \sigma_i \mathbf{u}_i \mathbf{v}_i^\top \mathbf{x} = \sum_{i=1}^r \sigma_i c_i \mathbf{u}_i$$

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- Note that $\sum_{i=1}^r \sigma_i \mathbf{u}_i \mathbf{v}_i^\top$ can be written in matrix form as

$$\mathbf{A} = \sum_{i=1}^r \sigma_i \mathbf{u}_i \mathbf{v}_i^\top = \mathbf{U} \mathbf{D} \mathbf{V}^\top$$

where $\mathbf{u}_i$ is the i^{th} column of $\mathbf{U}$, matrix $\mathbf{D}$ is diagonal with σ_i 's on the diagonal and $\mathbf{v}_i$ is the i^{th} column of $\mathbf{V}$:

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- For all $k \leq r$, we let $\mathbf{A}_k = \sum_{i=1}^k \sigma_i \mathbf{u}_i \mathbf{v}_i^\top$ be the truncation of the sum to the first k terms, corresponding to k largest singular values σ_i .
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- We define the 2-norm $\|\mathbf{A}\|_2$ of a matrix $\mathbf{A}$ as

$$\|\mathbf{A}\|_2 = \max_{|\mathbf{v}|=1} |\mathbf{A}\mathbf{v}| = \sigma_1.$$

- One can prove the following important facts about $\mathbf{A}_k$:
- **Theorem:** For every matrix $\mathbf{A} = \sum_{i=1}^r \sigma_i \mathbf{u}_i \mathbf{v}_i^\top$ of rank r and every $k < r$,

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Also, for every matrix $\mathbf{B}$ of rank at most k

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$$\|\mathbf{A} - \mathbf{A}_k\|_2 = \sigma_{k+1}$$

and for every matrix $\mathbf{B}$ of rank at most k

$$\|\mathbf{A} - \mathbf{A}_k\|_2 \leq \|\mathbf{A} - \mathbf{B}\|_2.$$

Thus, also in the sense of the 2-norm, matrix $\mathbf{A}_k$ is the best rank k approximation of matrix $\mathbf{A}$.

- **Theorem:** $\mathbf{A}\mathbf{v}_i = \sigma_i\mathbf{u}_i$ and $\mathbf{A}^\top\mathbf{u}_i = \sigma_i\mathbf{v}_i$.

- **Proof:** $\mathbf{A}\mathbf{v}_i = \sigma_i\mathbf{u}_i$ by definition. Let $\mathbf{A} = \sum_{i=1}^r \sigma_i\mathbf{u}_i\mathbf{v}_i^\top$, then

$$\mathbf{A}^\top = \sum_{i=1}^k \sigma_i\mathbf{v}_i\mathbf{u}_i^\top$$

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- Clearly the theorem implies that $\mathbf{A}\mathbf{v}_i/\|\mathbf{A}\mathbf{v}_i\| = \mathbf{u}_i$ and $\mathbf{A}^\top\mathbf{u}_i/\|\mathbf{A}^\top\mathbf{u}_i\| = \mathbf{v}_i$
- Note that

$$\mathbf{A}^\top\mathbf{A} = \left(\sum_{i=1}^r \sigma_i\mathbf{v}_i\mathbf{u}_i^\top \right) \left(\sum_{j=1}^r \sigma_j\mathbf{u}_j\mathbf{v}_j^\top \right) = \sum_{i,j=1}^r \sigma_i\sigma_j\mathbf{v}_i\mathbf{u}_i^\top\mathbf{u}_j\mathbf{v}_j^\top = \sum_{i=1}^r \sigma_i^2\mathbf{v}_i\mathbf{v}_i^\top$$

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- This means that σ_i^2 is an eigenvalue of $\mathbf{A}^\top \mathbf{A}$ and that $\mathbf{v}_j$ its eigenvector.
- Also,

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- But, since σ_1 is the largest singular value, σ_1^{2k} dominates all other σ_j^{2k} for $j > 1$.
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- As before, every vector $\mathbf{x}$ can be represented as $\mathbf{x} = \sum_{i=1}^r c_i \mathbf{v}_i + \mathbf{y}$ where $\mathbf{y}$ is orthogonal to all $\mathbf{v}_i$.
- Then

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- This implies

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- This suggests the following algorithm for finding $\mathbf{v}_1$ and $\mathbf{u}_1$:
 - 1 Start with a random vector $\mathbf{x}$ and normalise it so that $|\mathbf{x}| = 1$;
 - 2 Compute $\mathbf{z}_1 = \mathbf{A}\mathbf{x}$; compute $\mathbf{z}_2 = \mathbf{A}^T \mathbf{z}_1$; let $\rho = |\mathbf{z}_2|/|\mathbf{x}|$ and $\mathbf{x} = \mathbf{z}_2$
 - 3 Repeat (2) until the difference between ρ 's obtained in two consecutive iterations is smaller than a small threshold ε .
 - 4 Set $\mathbf{v}_1 = \mathbf{z}_2/|\mathbf{z}_2|$;
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Applications of SVD: ranking webpages

- We describe one of the first proposals for ranking webpages according to their significance, called HITS algorithm, proposed by one of the authors of your textbook, Jon Kleinberg.
- Let $\mathbf{A}$ be the adjacency matrix of the webpages; thus, $\mathbf{A}_{ij} = 1$ if page P_i has a hyperlink to page P_j and 0 otherwise.
- Some web pages, called authorities, are prominent sources for information on a given topic. Some other pages, called hubs, identify the authorities on a topic.
- Authority pages those that are pointed to by many hub pages and hub pages are those which point to many authority pages.
- Clearly, this is a circular definition, so we have to assign the authority ranks $\mathbf{a}$ of webpages, as well as their hub ranks $\mathbf{h}$ in a simultaneous manner.
- Note that given a vector of hub ratings $\mathbf{h}$ we can compute the authority ratings $\mathbf{a}$ of websites, by letting the j^{th} coordinate of $\mathbf{a}$, i.e., a_j to be the sum of the hub ranks of all webpages i which point to j , i.e., as $a_j = \sum_i \mathbf{A}_{ij} h_i$.
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- We describe one of the first proposals for ranking webpages according to their significance, called HITS algorithm, proposed by one of the authors of your textbook, Jon Kleinberg.
- Let $\mathbf{A}$ be the adjacency matrix of the webpages; thus, $\mathbf{A}_{ij} = 1$ if page P_i has a hyperlink to page P_j and 0 otherwise.
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- This property is satisfied by the right and the left singular vectors and we pick the ones corresponding to the largest singular value, i.e., $\mathbf{a} = \mathbf{v}_1$ and $\mathbf{h} = \mathbf{u}_1$.
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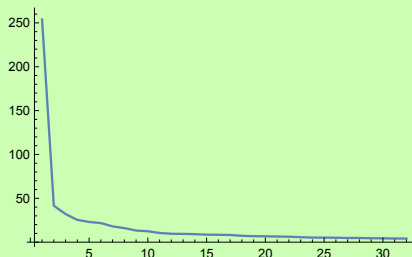
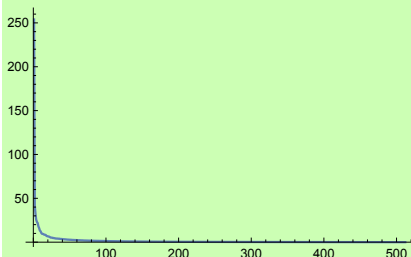
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Singular Values of Lena image



- Singular values of the photo of Lena represented as a matrix $\mathbf{A}$ of size 512×512
- On the left: all 512 values
- On the right: 32 largest values
- Next page: matrices $\mathbf{A}_k = \sum_{i=1}^k \sigma_i \mathbf{u}_i \mathbf{v}_i^T$ for $k = 16, 32, 64, 128, 256, 512$



A_{16} A_{128} A_{32} A_{256} A_{64} A_{512} 